In this journal entry, I will talk about my experiences and findings from module 8. In this module, we learned about underfitting and overfitting learning curves, how they work, how to see which type of learning curve it is, and what the best fit model will look like. This module taught us about an underfitting learning curve vs. an overfitting curve vs a good fit. This module taught us how each of these works, and I will reflect on how each of these impacted my learning.

The first thing I learned about in this module is different levels of polynomial regression models. And what each one represents when looking at the curve with the dataset as well. The first polynomial shown is to the 1st degree; this would be an example of underfitting a model, as the regression model is incredibly simple, only fitting one line, and it is not a great predictor. The next polynomial is displayed to the 4th degree. This is a great fit model, as this is a bit more complex than just a straight line but not too complex to be moved so much by little changes in the data; this is what you want to aim for in a model. Finally, there is the 15th degree polynomial; this was a great example of overfitting, as this curve showed signs of change no matter how small the change in the dataset was. However, this made me question how data scientists choose the right degree, since you may find a range of degrees that work well. How would you compare these, especially if it's a larger dataset with fewer individual data points, to see which degree would work better? Then I thought about the cross validation method mentioned later in the lab: if one degree does better on one train test split but a different degree does better on another split, that could become an issue in picking the right one. This makes me wonder how data scientists pick between options like these that seem to be similar now, but could have a major impact later on.

This leads me to the next thing this lab taught me about: the cross validation method. Before this lab, we used the standard 70 30 split method. This is the best and only method to test a machine learning model. This lab showed me the flaws of the train test split method. Firstly, the train test method only uses one part of the data for testing. This can be an issue, as if the part the machine is tested on is really easy or hard, it can skew the results of the actual accuracy of the model. Instead, the cross validation method will take 5 separate train and test data sets from a different 20% each time, so the average of all these scores will be much more accurate than just 1 train test split. However, you are running the train test split 5 times instead of just once, which means that this takes a lot more processing power than just a regular train test split, which is why this is only recommended if your dataset is under a million rows. This made me think about unbalanced datasets: would cross validation work there too, or because of the lopsided dataset it would become less effective than the train test. If that were so than the train test also could not be lopsided. I would feel that, especially for lopsided datasets that you would like to do cross validation, as this makes sure in a single train test that you're not missing anything.

Finally, this lab showed me the difference between the plot of underfitting and overfitting graphs. The first graph shows a very low score while having a small gap. This means that the underfitting model won't have much accuracy. This makes sense, as almost every dataset will curve and change, so a single straight line will not be able to follow the data points and will have low accuracy. While underfitting also has a small gap, this is because the model is not actually learning anything, so it can't memorize data. Meanwhile, overfitting will have a high score since it moves to every little change in the plot; it has great accuracy, but a major gap because instead of learning, it's just memorizing.

In conclusion, this module again taught me that every part of the machine learning process is just as important as the other and, you cannot make one without careful consideration at every step. This showed me that finding the right balance between too many polynomials and too little. Yet again, this module shows in machine learning that balance is key and is more important than anything else. This module showed me cross validation this was a new way to do something I thought was the best way. This just shows me that there is still much more to learn and, I hope to learn that in the next modules.